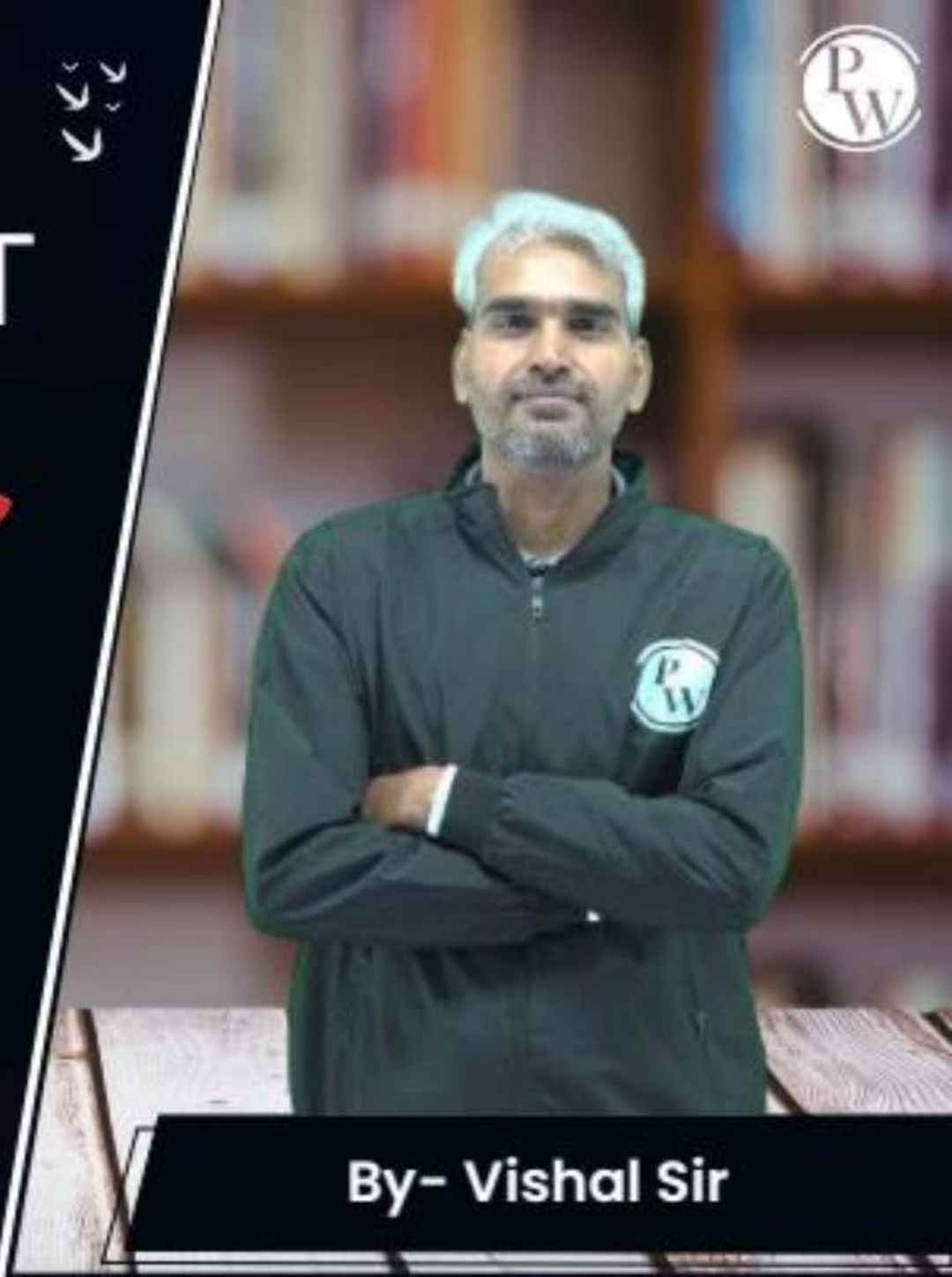
Computer Science & IT



Discrete Mathematics

Mathematical Logic

Lecture No. 03



By- Vishal Sir

Recap of Previous Lecture



Topic Logical implication

Topic Logical equivalence

Topic Argument and rules of inference

Slide

Topics to be Covered



Topic Rules of inference

Topic Proof by contradiction

Topic Conditional proof

Slide

H.W

Q. Simplify the following statement

$$\{\sim p \wedge (\sim q \wedge r)\} \vee \{(q \wedge r) \vee (p \wedge r)\}$$

$$\{(\sim p \wedge \sim q) \wedge r\} \vee \{(q \vee p) \wedge r\}$$

$$\{\sim (p \vee q) \wedge r\} \vee \{(p \vee q) \wedge r\}$$

$$\{\sim (p \vee q) \vee (p \vee q)\} \wedge r$$

$$T \wedge r$$

$$\equiv r$$

(a) True
(b) False
(c) p
(d) q
(e) r

H.W.

Q: Simplify the following Expression

$$[(P \vee Q) \wedge \sim\{\sim P \wedge (\sim Q \vee \sim R)\}] \vee [(\sim P \wedge \sim Q) \vee (\sim P \wedge \sim R)]$$

$$[(P \vee Q) \wedge \sim\{(\sim P \wedge \sim Q) \vee (\sim P \wedge \sim R)\}] \vee [\sim P \wedge (\sim Q \vee \sim R)]$$

$$[(P \vee Q) \wedge \{(P \vee Q) \wedge (P \vee R)\}] \vee \sim [P \vee (Q \wedge R)]$$

$$[\{(P \vee Q) \wedge (P \vee R)\}] \vee \sim [P \vee (Q \wedge R)]$$

$$\underbrace{[P \vee (Q \wedge R)]}_{X} \quad \vee \sim \underbrace{[P \vee (Q \wedge R)]}_{\sim X}$$

$$\equiv \text{True}$$

H.W. Q: Which of the following is/are valid?

(a) $(P \wedge Q)$ logically implies $(P \leftrightarrow Q)$

$T \wedge T \equiv T$; $T \leftrightarrow T \equiv T$

(b) $(P \leftrightarrow Q)$ logically implies $(P \rightarrow Q)$

$T \leftrightarrow T \equiv T$; $T \rightarrow T \equiv T$

$f \leftrightarrow f \equiv T$; $f \rightarrow f \equiv T$

$\therefore$ valid

(c) $(P \leftrightarrow Q)$ logically implies $(P \rightarrow \sim Q)$

$T \leftrightarrow T \equiv T$; $T \rightarrow f \equiv F$

$f \leftrightarrow f \equiv T$; $f \rightarrow T \equiv T$

$\therefore$ invalid

(d) Q logically implies $(P \rightarrow Q)$

T ; $\phi \rightarrow T \equiv T$

$\therefore$ valid

P logically implies Q is valid

if and only if

$P \rightarrow Q$ is a tautology

H.W. Q: Which of the following is/are tautology ?

(a) $(P \wedge Q) \longrightarrow (P \longleftrightarrow Q)$

(b) $(P \longleftrightarrow Q) \longrightarrow (P \longrightarrow Q)$

(c) $(P \longleftrightarrow Q) \longrightarrow (P \longrightarrow \sim Q)$

(d) $Q \longrightarrow (P \longrightarrow Q)$

H.W.

Q: Which of the following is/are tautology?

(a) $(a \vee b) \rightarrow (b \wedge c)$

$T \vee F \equiv T \longrightarrow F \wedge \emptyset \equiv F$

(b) $(a \wedge b) \rightarrow (b \vee c)$

$\emptyset \wedge F \equiv F \qquad F \vee F$

(c) $(a \vee b) \rightarrow (b \rightarrow c)$

$\emptyset \vee T \equiv T \qquad T \rightarrow F \equiv F$

(d) $(a \rightarrow b) \rightarrow (b \rightarrow c)$

$\emptyset \rightarrow T \equiv T \qquad T \rightarrow F \equiv F$

H.W

Q: Consider the following statements

- S_1: $\{(A \wedge B) \longrightarrow C\} \equiv \{(A \rightarrow C) \wedge (B \rightarrow C)\}$
- S_2: $\{(A \vee B) \longrightarrow C\} \equiv \{(A \rightarrow C) \vee (B \rightarrow C)\}$

then which of the following is true ?

(a) Both S_1 & S_2 are valid

(b) Only S_1 is valid

(c) Only S_2 is valid

(d) Neither S_1 nor S_2 is valid.

H.W

Q: Consider the following statements

S_1: $\{(A \wedge B) \rightarrow C\} \equiv \{(A \rightarrow C) \wedge (B \rightarrow C)\}$

S_2: $\{(A \vee B) \rightarrow C\} \equiv \{(A \rightarrow C) \vee (B \rightarrow C)\}$

→ L.H.S = $\{(A \wedge B) \rightarrow C\}$

$\{\sim(A \wedge B) \vee C\}$

$\{(\sim A \vee \sim B) \vee C\}$

$\{(\sim A \vee \sim B) \vee (C \vee C)\}$

$X \vee X = X$

$\{(\sim A \vee C) \vee (\sim B \vee C)\} = \{(A \rightarrow C) \vee (B \rightarrow C)\} \neq$ RHS

Q: Consider the following statements

S_1: $\{(A \wedge B) \longrightarrow C\} \not\equiv \{(A \rightarrow C) \wedge (B \rightarrow C)\}$

$$\{(A \wedge B) \longrightarrow C\} \equiv \{(A \rightarrow C) \vee (B \rightarrow C)\}$$

$\{(A \wedge B) \longrightarrow C\} \not\Longrightarrow \{(A \rightarrow C) \wedge (B \rightarrow C)\}$

invalid

$\{(A \wedge B) \longrightarrow C\} \Longleftarrow \{(A \rightarrow C) \wedge (B \rightarrow C)\}$

Valid

H.W

Q: Consider the following statements

- S_1: $\{(A \wedge B) \rightarrow C\} \equiv \{(A \rightarrow C) \wedge (B \rightarrow C)\}$
- S_2: $\{(A \vee B) \rightarrow C\} \equiv \{(A \rightarrow C) \vee (B \rightarrow C)\}$

↳ LHS: $\{(A \vee B) \rightarrow C\}$

$\equiv \{\sim(A \vee B) \vee C\}$

$\equiv \{(\sim A \wedge \sim B) \vee C\}$

$\equiv \{(\sim A \vee C) \wedge (\sim B \vee C)\} \equiv (A \rightarrow C) \wedge (B \rightarrow C) \not\equiv$ RHS.

H.W Q: Consider the following statements

- S_2: $\{(A \vee B) \rightarrow C\} \not\equiv \{(A \rightarrow C) \vee (B \rightarrow C)\}$

$$\{(A \vee B) \rightarrow C\} \equiv \{(A \rightarrow C) \wedge (B \rightarrow C)\}$$

$$\{(A \vee B) \rightarrow C\} \Longrightarrow \{(A \rightarrow C) \vee (B \rightarrow C)\}$$ ✓ Valid

$$\{(A \vee B) \rightarrow C\} \not\Longleftarrow \{(A \rightarrow C) \vee (B \rightarrow C)\}$$ Invalid

H.W

Q: Consider the following statements

- S_1: $\{(A \wedge B) \longrightarrow C\} \equiv \{(A \rightarrow C) \wedge (B \rightarrow C)\}$
- S_2: $\{(A \vee B) \longrightarrow C\} \equiv \{(A \rightarrow C) \vee (B \rightarrow C)\}$

then which of the following is true ?

(a) Both S_1 & S_2 are valid

(b) Only S_1 is valid

(c) Only S_2 is valid

(d) Neither S_1 nor S_2 is valid.

H.W. Q: Which of the following is are true ?

(a) $\{a \to (b \vee c)\} \equiv \{(a \wedge \sim b) \to c\}$

LHS $= \{a \to (b \vee c)\} \equiv \{\sim a \vee (b \vee c)\} \equiv \{(\sim a \vee b) \vee c\} \equiv \{\sim(a \wedge \sim b) \vee c\}$
$\equiv (a \wedge \sim b) \to c \equiv$ RHS

(b) $\{(P \to Q) \wedge (R \to Q)\} \equiv \{(P \vee R) \to Q\}$

(c) $\{(P \to Q) \vee (R \to Q)\} \equiv \{(P \wedge R) \to Q\}$

} Valid from previous question

(d) All of the above

H.W.

Q. Consider the following Propositional statement.

$$\{P \rightarrow (Q \vee R)\} \longrightarrow \{(P \wedge Q) \rightarrow R\}$$

Above Propositional statement is.

- (a) Satisfiable but not valid {i.e. Contingency}
- (b) Valid
- (c) Contradiction

H.W. Q. Consider the following Propositional statement.

$$\{P \rightarrow (Q \vee R)\} \longrightarrow \{(P \wedge Q) \rightarrow R\}$$

If $R = \text{True}$ then it is True
∴ Whole implication will be true
Hence, True for at least one case

In this case
what will happen with LHS

$$P \quad Q \quad R$$
$$T \rightarrow (T \vee F) \equiv T$$

If $P = Q = T$ & $R = F$ then
in that case it will be false

When RHS is false then
LHS can be true ∴ implication can be false as well ∴ Not valid

H.W. Q. Consider the following Propositional statement.

$$\{P \rightarrow (Q \vee R)\} \longrightarrow \{(P \wedge Q) \rightarrow R\}$$

Above Propositional statement is.

(a) Satisfiable but not valid { i.e. Contingency }

(b) Valid

(c) Contradiction

Statement S_1: If it rains today then Cricket match is not played

Statement S_2: Cricket match is not played

Conclusion: It rained today

→ it is an argument

→ it may be valid
or
it may be invalid

Our job is to check whether the argument is valid or invalid



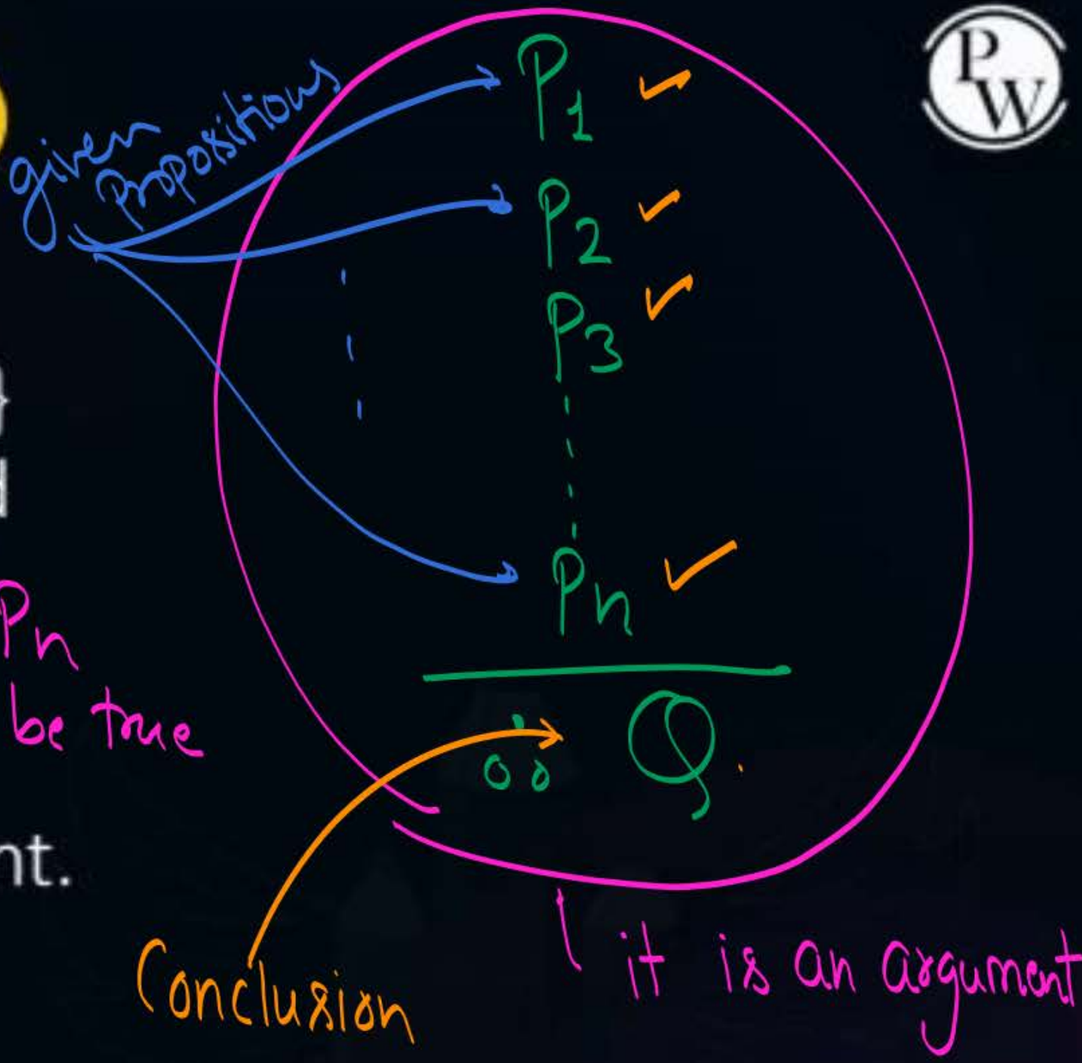
Topic : Argument / Inference

The statement that,

A set of premises { P1, P2, P3,.....,Pn} yields another proposition Q is called an argument.

The argument is, that whenever $P_1, P_2, \ldots P_n$ are true then Conclusion Q is guaranteed to be true

Q is called conclusion of the argument.





Topic : Argument / Inference

- Argument may be valid or invalid.
- The process of reasoning whether the argument is valid or invalid is called inference.
- If conclusion Q can be inferred from the set of premises by applying some rules of inference and equivalences, then argument is said to be valid otherwise invalid.



Topic : Rules of inference

Any valid reasoning is rule of inference.

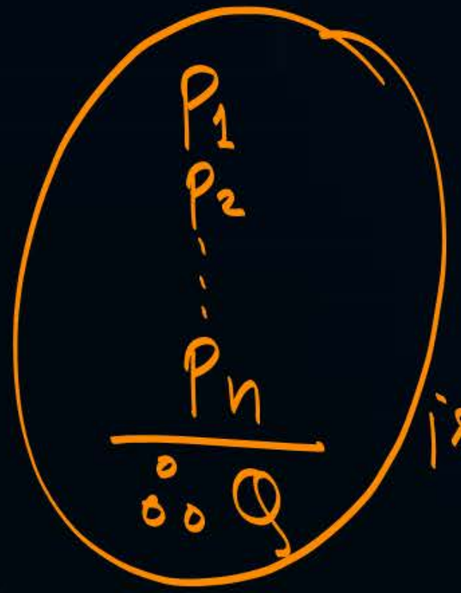


Topic : Argument / Inference

Following statements are equivalent,

- Argument {P1, P2, P3,.....,Pn} $\vdash Q$ is valid. (yields)
- {P1, P2, P3,.....,Pn} Logically implies Q is valid
- {P1 ∧ P2 ∧ P3 ∧ ..... ∧ Pn} → Q is a tautology
- Conclusion Q follows from the premises $\{P_1, P_2, \cdots P_n\}$

$\equiv$



is valid

Rules of inference



Topic : Rules of inference

1. Simplification

$$\frac{P \wedge Q}{\therefore P}$$ is valid

Similarly $$\frac{P \wedge Q}{\therefore Q}$$ is vali

- P follows from $(P \wedge Q)$
- $(P \wedge Q)$ logically implies Q is valid
- $(P \wedge Q) \rightarrow P$ is a tautology
- $(P \wedge Q) \vdash P$ is valid



Topic : Rules of inference

2. Addition

$$\frac{P}{\therefore P \vee Q}$$ is valid

- $P \rightarrow (P \vee Q)$ is a tautology
- $(P \vee Q)$ follows from P



Topic : Rules of inference

2. Addition

$$\frac{P}{\therefore P \vee Q}$$ is valid

→ Addition can be performed using disjunction (not using Conjunction)

i.e $\frac{P}{\therefore P \wedge Q}$ is invalid

→ Simplification can be performed w.r.t. Conjunction Only

i.e. $\frac{P \vee Q}{\therefore P}$ is invalid



Topic : Rules of inference

3. Conjunction

(i) P

(ii) Q

$\therefore P \wedge Q$ is valid



Topic : Rules of inference

4. Disjunctive Syllogism

$$\frac{p \vee q \quad \sim p}{\therefore q}$$ is Valid

$$\frac{p \vee q \quad \sim q}{\therefore p}$$ is Valid



Topic : Rules of inference

5. Conjunctive Syllogism

$\sim(P \wedge Q)$ $\Rightarrow$ $(P \wedge Q)$ is false

P $\Rightarrow$ P is true

$\therefore \sim Q$ is valid $\Rightarrow$ $\therefore Q$ is false



Topic : Rules of inference

6. Modus Ponens

$$\frac{p \rightarrow q \qquad p}{\therefore\ q}$$

is valid

7. Fallacy of affirming the consequent

Mistake

$$\frac{p \rightarrow q \qquad q}{\therefore\ p}$$

is invalid



Topic : Rules of inference

8. Modus Tollen's

$$\begin{array}{l} p \rightarrow q \\ \sim q \\ \hline \therefore \sim p \end{array}$$

is valid

9. Fallacy of denying the antecedent

$$\begin{array}{l} p \rightarrow q \\ \sim p \\ \hline \therefore \sim q \end{array}$$

is invalid



Topic : Rules of inference

10. Transitivity

$$\frac{P \rightarrow Q \qquad Q \rightarrow R}{\therefore P \rightarrow R}$$ is Valid



Topic : Rules of inference

11. Dilemma

$P \vee Q$: At least one of P or Q is definitely True

$P \rightarrow R$

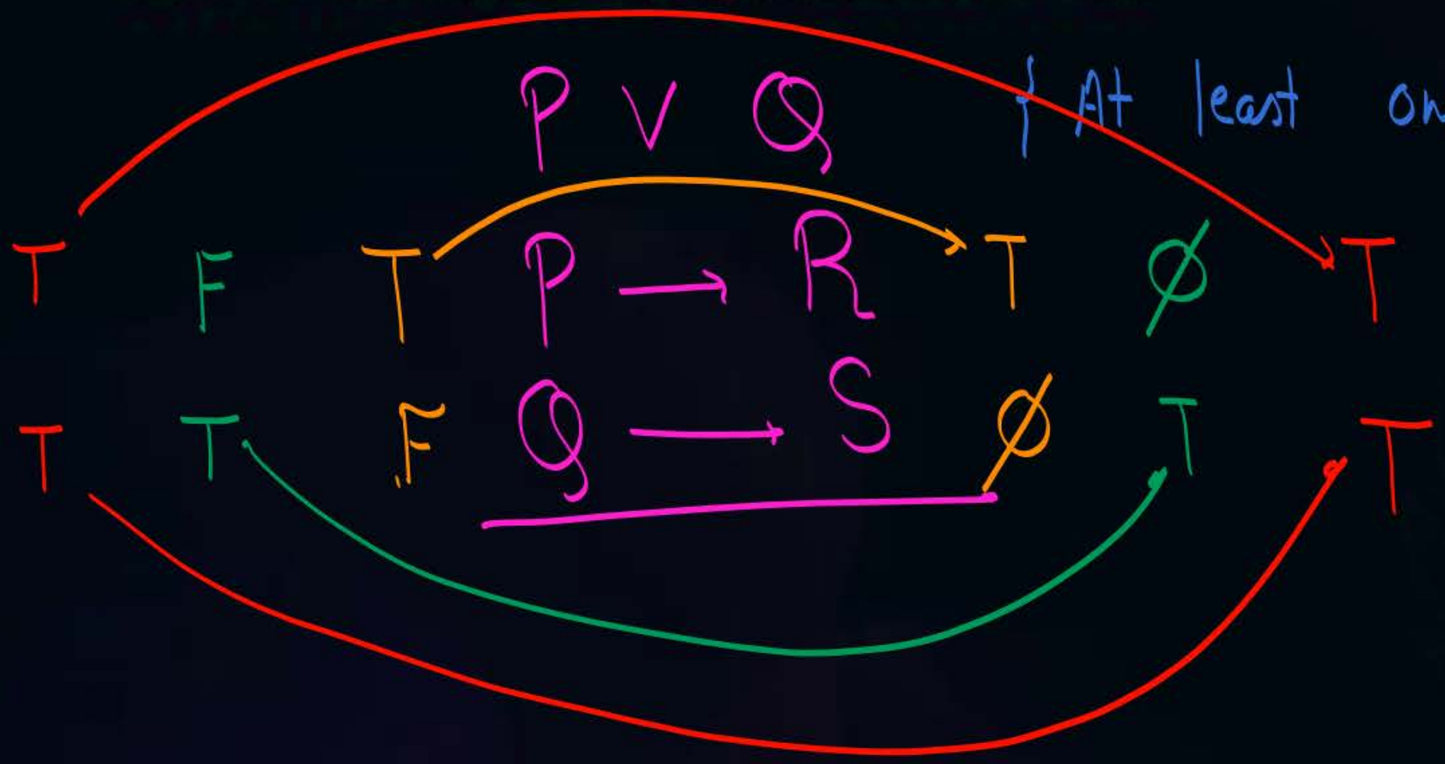
$Q \rightarrow R$

$\therefore R$ is valid



Topic : Rules of inference

12. Constructive Dilemma



At least one of P or Q is definitely True



Topic : Rules of inference

12. Constructive Dilemma

$P \vee Q$ { At least one of P or Q is definitely True

$P \rightarrow R$

$Q \rightarrow S$

$\therefore R \vee S$ is Valid



Topic : Rules of inference

13. Destructive Dilemma

$P \rightarrow R$

$Q \rightarrow S$

$\sim R \vee \sim S$: At least one of R or S is definitely false

---

$\therefore \sim P \vee \sim Q$: At least one of P or Q is definitely false

is valid



Topic : Rules of inference

14. Resolution

$$p \vee q$$

$$\sim p \vee r$$

$$\therefore q \vee r$$ is valid

$p \vee q$

$\sim p \vee r$

$\therefore q \vee r$

Exactly one of p or ~p is definitely false

$\therefore$ At least one of q or r must be true



Topic : Rules of inference

15. Non Sequitur

$$\frac{p \quad q \rightarrow r}{\therefore \quad r}$$

is invalid

Q:- Which of the following is/are true?

(a) Conclusio R follows from the premises.

$\{P \rightarrow (Q \rightarrow R),\ (P \wedge Q)\}$

(b) $\{P \rightarrow (R \rightarrow S),\ \sim R \rightarrow \sim P,\ P\}$ logically implies S

(c) $\{(P \vee Q),\ \sim P\} \vdash (P \wedge Q)$ is valid

(d) $\{(\sim T \rightarrow \sim R) \wedge \sim S \wedge (T \rightarrow W) \wedge (R \vee S)\} \rightarrow W$ is a tautology

Following statement are equivalent

① Argument $\left\{\begin{array}{l} P_1 \\ P_2 \\ \vdots \\ P_n \\ \hline \therefore\ Q \end{array}\right\}$ is valid

② Q follows from the premises $\{P_1, P_2, \ldots, P_n\}$

③ $\{P_1, P_2, \ldots, P_n\} \vdash Q$ (yields) is valid

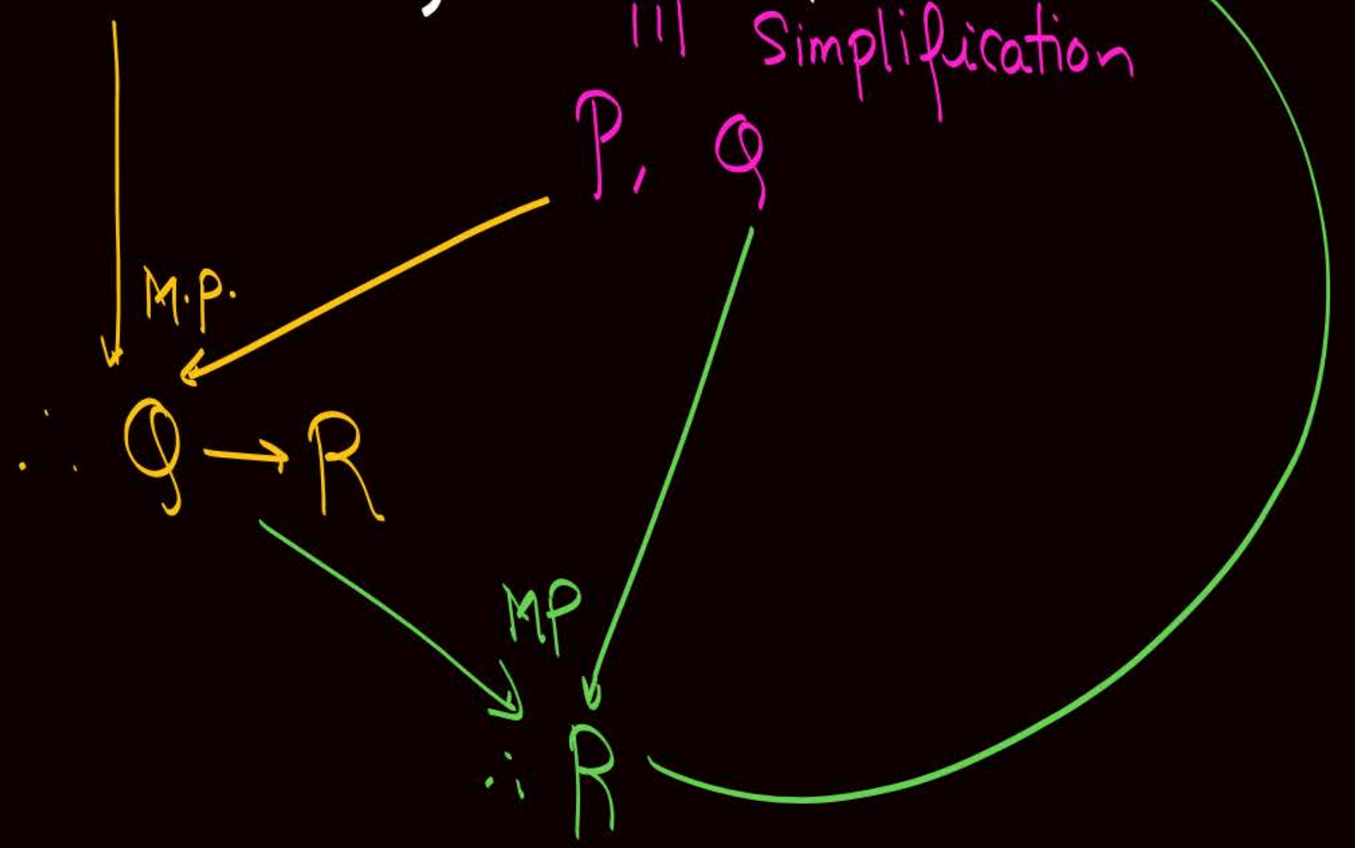
④ $\{P_1 \wedge P_2 \wedge \ldots \wedge P_n\}$ logically implies Q is true

⑤ $\{P_1 \wedge P_2 \wedge \ldots \wedge P_n\} \rightarrow Q$ is a tautology.

Q:- Which of the following is/are true?

(a) Conclusio R follows from the premises.

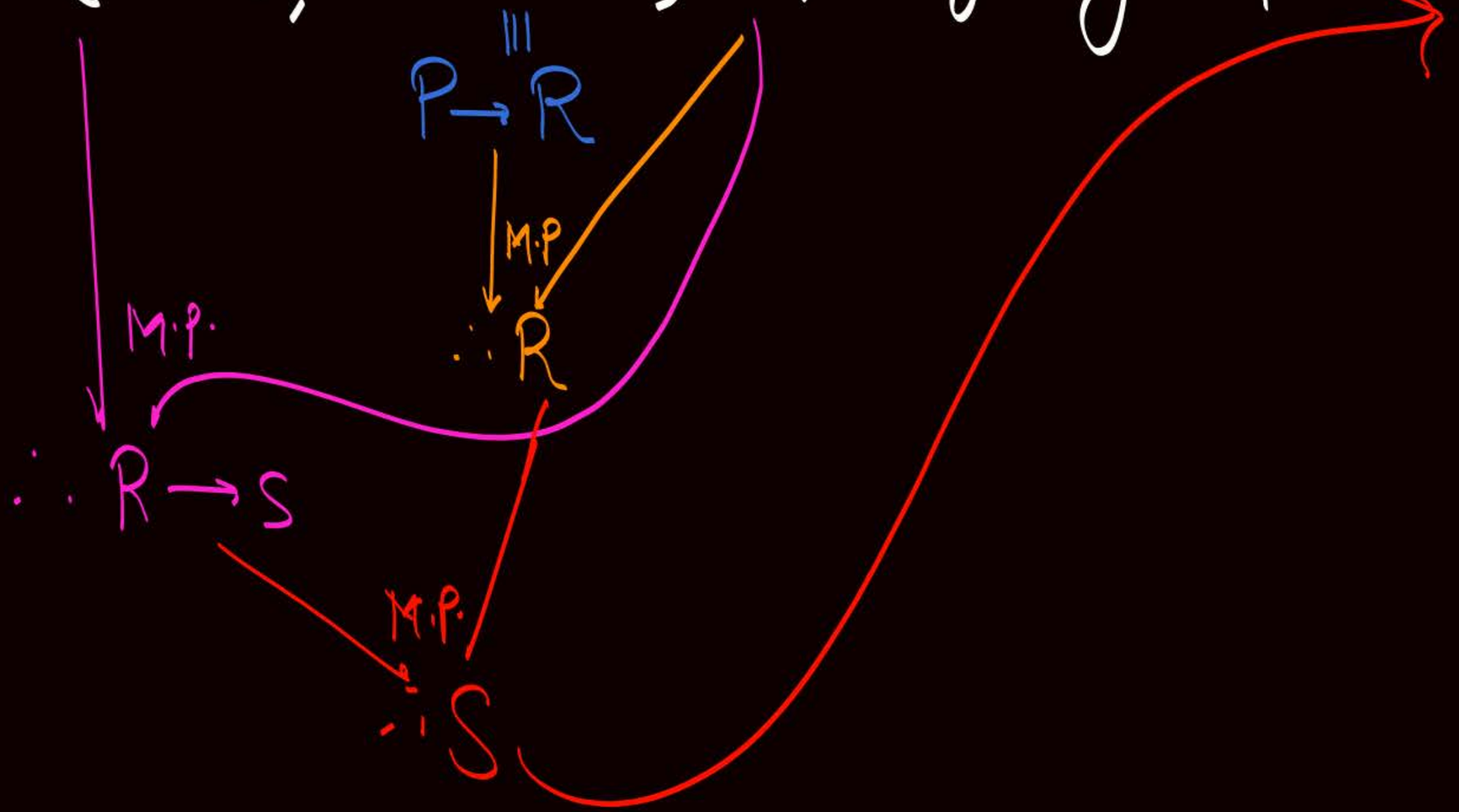
$\{P \rightarrow (Q \rightarrow R),\ (P \wedge Q)\}$



$\therefore$ R follows from given Premises.

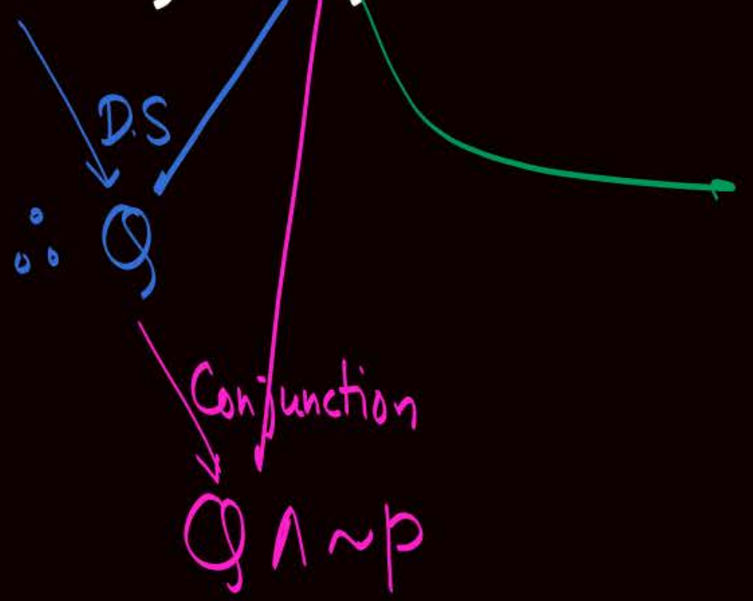
Q:- Which of the following is/are true?

(b) $\{P \to (R \to S),\ \sim R \to \sim P,\ P\}$ logically implies S



Q:- Which of the following is/are true?

© $\{(P \vee Q), \sim P\} \vdash (P \wedge Q)$ is valid



if ~p is true

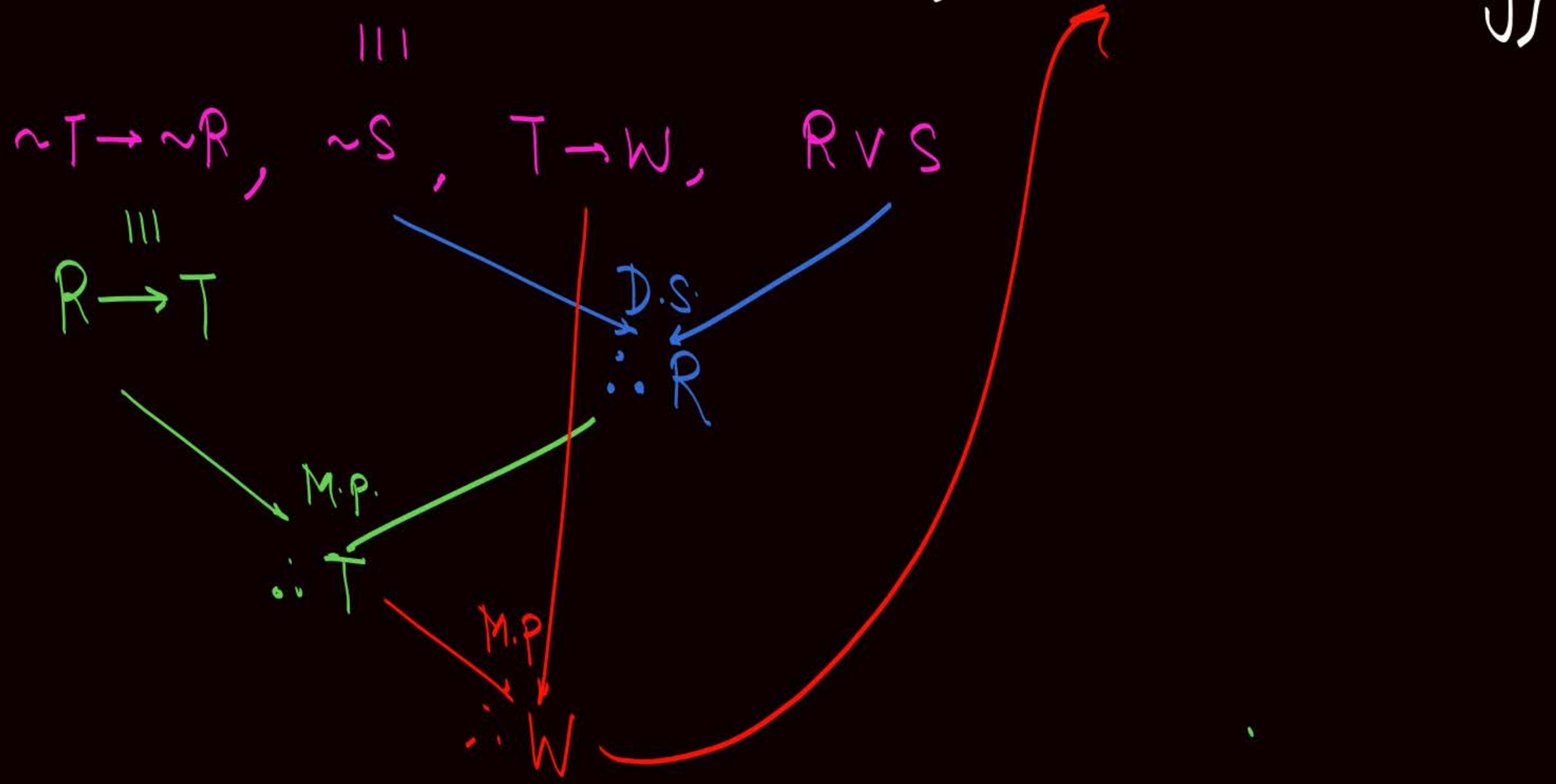
i.e, if 'p' is false,

then "p ∧ Q" can never be true

∴ invalid

Q:- Which of the following is/are true?

(d) $\{(\sim T \to \sim R) \wedge \sim S \wedge (T \to W) \wedge (R \vee S)\} \to W$ is a tautology



Q:- Which of the following is/are true?

(a) Conclusio R follows from the premises.

$\{P \rightarrow (Q \rightarrow R),\ (P \wedge Q)\}$

(b) $\{P \rightarrow (R \rightarrow S),\ \sim R \rightarrow \sim P,\ P\}$ logically implies S

(c) $\{(P \vee Q),\ \sim P\} \vdash (P \wedge Q)$ is valid

(d) $\{(\sim T \rightarrow \sim R) \wedge \sim S \wedge (T \rightarrow W) \wedge (R \vee S)\} \rightarrow W$ is a tautology

H.W.

Q:- Check whether the following argument is valid or invalid

∴ If today is Vishal's Bday, then today is 1st April

∴ Today is 1st April

∴ Today is Vishal's B'day

Let p = Today is Vishal's Bday

q = Today is 1st April

Q:- Check whether the following argument is valid or invalid

HW:

- If today is Vishal's Bday, then today is 1st April
- Today is not 1st April

∴ Today is not Vishal's B'day

p = Today is Vishal's Bday

~p = Today is not Vishal's Bday

q = Today is 1st April

~q = Today is not 1st April

* Q:. Check whether the following argument is valid or invalid

H.W.

$\{ p \rightarrow q , \sim(p \wedge q) \}$ logically implies $\sim p$



2 mins Summary

Topic Rules of inference

Topic Proof by contradiction

Topic Conditional proof

Slide

THANK - YOU